Dynamics of phenotypically structured populations in time-periodic environments:

A Hamilton-Jacobi approach

Susely Figueroa Iglesias[1] Sepideh Mirrahimi[2]

[1] Institut de Mathématiques de Toulouse, Paul Sabatier University
Susely.Figueroa@math.univ-toulouse.fr

[2] Institut de Mathématiques de Toulouse, Paul Sabatier University
Sepideh.Mirrahimi@math.univ-toulouse.fr

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Biological motivation

Several populations of bacterial pathogen *Serratia marcescens* were kept in environments with constant or fluctuating temperature for several weeks and then measured their growth rates in different environments. The population evolved in a periodic environment outperforms the population evolved in a constant environment.

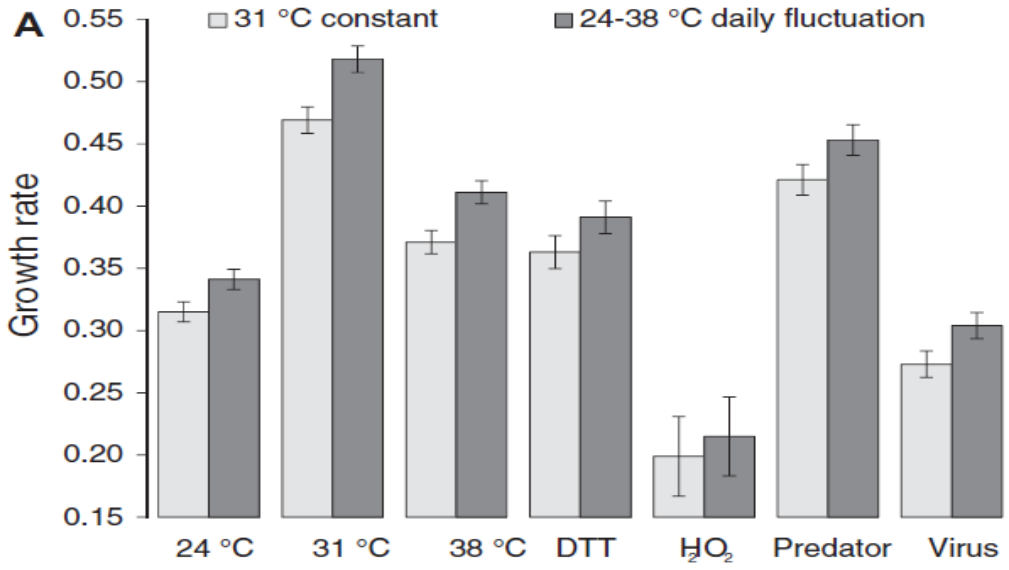

Figure 1: Average maximal growth rate of Serratia marcescens-bacteria evolved either in constant $31°$C or in daily fluctuating temperature between $24°$C and $38°$C, (Figure from Ketola et al. 2013)

Motivations and Questions

We study the dynamics of a phenotypically structured population under the effect of mutations-selection in a fluctuating environment.

- What will be the impact of the variations of the environment on the population's phenotypical distribution?
- In which situations a population evolved in a periodic environment outperforms the population evolved in a constant environment ?

Periodic variations: seasonal effects, prescription of drugs, (antibiotic resistance)...

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Outline

1. Model and previous works
2. Asymptotics as the mutations vanish
3. Biological Application

Section 1

Model and previous works

Model and Notation

A parabolic integro-differential model with $(t,x) \in [0,+\infty) \times \mathbb{R}^d$ with non-local competition

$$\begin{cases} \partial_t n - \overbrace{\varepsilon^2 \Delta n}^{\text{Mutation}} = n\big[\overbrace{a(e(t),x)}^{\text{Growth}} - \overbrace{\rho(t)}^{\text{Competition}}\big], \\ \rho(t) = \displaystyle\int_{\mathbb{R}^d} n(t,x)\,dx, \\ n(t=0,x) = n_0(x). \end{cases} \tag{P}$$

- x : phenotypic trait
- $n(t,x)$: density of trait x
- $\rho(t)$: total size
- $a(e(t),x)$: net growth rate
- $e(t)$: environmental state
- ε^2: effect of mutations

Assumptions

- $e(t)$ is $T-$periodic
- $a(t,x) \in L^\infty(\mathbb{R}_+, C^3(\mathbb{R}))$ is negative for x large
- $\exists!\ x_m : \max_{x\in\mathbb{R}^d} \overline{a}(x) = \overline{a}(x_m) > 0$ for $\overline{a}(x) = \frac{1}{T}\int_0^T a(t,x)\,dt$
- $n_0(x) \leq e^{C_1 - C_2|x|}$

Convergence to a periodic solution

Proposition (Fl., Mirrahimi 2018)

For small values of ε we have

$$|\rho(t) - \rho_\varepsilon(t)| \to 0, \quad \|n(t,\cdot) - n_\varepsilon(t,\cdot)\|_{L^\infty} \to 0, \quad \text{as } t \to \infty,$$

for $(n_\varepsilon, \rho_\varepsilon)$ the unique periodic solution of

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Goal: Study $(n_\varepsilon, \rho_\varepsilon)$ asymptotically as $\varepsilon \to 0$
Tool: Hopf-Cole transformation: $n_\varepsilon = \frac{1}{(2\pi\varepsilon)^{d/2}} \exp\left(\frac{u_\varepsilon}{\varepsilon}\right)$.
Expected asymptotic expansion: $u_\varepsilon(t,x) = u + \varepsilon v + O(\varepsilon^2)$.

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Previous Works

- *Works considering* ***oscillating environments***
  - Lorenzi et al. (2015)
  - Mirrahimi-Perthame-Souganidis (2015)
  - Almeida et al. (2019)

- *References using* ***asymptotic approach*** *based on Hamiton-Jacobi equations*
  - **on propagation phenomena**:
    Freidlin, Evans, Souganidis, Barles,...
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Section 2

Asymptotics as the mutations vanish

The Main Result

Theorem (Fl., Mirrahimi 2018)

(i) *As* $\varepsilon \to 0$*:*

$$n_\varepsilon(t,x) - \widetilde{\varrho}(t)\delta(x - x_m) \rightharpoonup 0,$$

where $\widetilde{\varrho}(t)$ *is the unique periodic solution of*

$$\frac{d\widetilde{\varrho}}{dt} = \widetilde{\varrho}[a(e(t), x_m) - \widetilde{\varrho}].$$

(ii) *As* $\varepsilon \to 0$, $u_\varepsilon(t,x) = \varepsilon \ln n_\varepsilon(t,x)$ *converges locally uniformly to* $u(x)$*, the unique viscosity solution of:*

$$\begin{cases} -|\nabla u|^2 = \overline{a}(x) - \overline{a}(x_m), & x \in \mathbb{R}^d, \\ \max\limits_{x\in\mathbb{R}^d} u(x) = u(x_m) = 0. \end{cases}$$

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Heuristics

Let $u_\varepsilon = \varepsilon \ln n_\varepsilon$, then for $(t,x) \in [0,+\infty) \times \mathbb{R}^d$:

$$\frac{1}{\varepsilon}\partial_t u_\varepsilon - \varepsilon \Delta u_\varepsilon = |\nabla u_\varepsilon|^2 + a(t,x) - \rho_\varepsilon(t).$$

Formally asymptotic expansion for u_ε and ρ_ε with T-periodic coefficients:

$$u_\varepsilon(t,x) = u(t,x) + \varepsilon v(t,x) + \varepsilon^2 w(t,x) + o(\varepsilon^2),$$

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$$\Downarrow$$

$$n_\varepsilon(t,x) \approx \frac{1}{(2\pi\varepsilon)d/2} e^{\frac{u(x)}{\varepsilon} + v(t,x) + \varepsilon w(t,x)}.$$

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Heuristics

Equation for $u(x)$:

$$-|\nabla u|^2 = \frac{1}{T}\int_0^T (a(t,x) - \rho(t))\ dt,$$

System equations for $v(t,x)$:

$$\begin{cases} \partial_t v &= a(t,x) - \overline{a}(x) - \rho(t) + \overline{\rho}, \\ -\Delta u &= \dfrac{2\nabla u}{T}\displaystyle\int_0^T \nabla v dt - \Delta u(x_m). \end{cases}$$

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Regularity estimates

Lemma (Eigenvalue problem)

The periodic solution n_ε can be written

$$n_\varepsilon(t,x) = \frac{p_\varepsilon(t,x)}{\int_{\mathbb{R}^d} p_\varepsilon(t,x)\,dx}\hat{\rho}_\varepsilon(t)$$

where p_ε is the solution of the eigenproblem

$$\begin{cases} \partial_t p_\varepsilon(t,x) - \varepsilon^2 \Delta p_\varepsilon(t,x) = p_\varepsilon(t,x)[a(e(t),x) - \lambda_\varepsilon], \\ 0 < p_\varepsilon(t,\cdot) = p_\varepsilon(t+T,\cdot) \end{cases}$$

which satisfies

$$p_\varepsilon(t,x) \leq \|p_\varepsilon\|_{L^\infty} e^{-\frac{1}{\varepsilon}\sqrt{\delta}(|x|-R_0)}, \quad \forall (t,x) \in [0,+\infty) \times \mathbb{R}^d.$$

Difficulty: $T-$periodicity $\Rightarrow$ no information on the initial conditions.

Regularity estimates 2

Theorem (Regularity of u_ε)

Let $u_\varepsilon = \varepsilon \ln n_\varepsilon(t,x)$, *then* $\forall (t,x) \in [0,+\infty) \times \mathbb{R}^d$

$$-A_1|x|^2 - A_2 \leq u_\varepsilon(t,x) \leq A_3 - A_4|x|,$$

$$|\nabla u_\varepsilon| \leq C,$$

$$|u_\varepsilon(t,x) - u_\varepsilon(s,x)| \to 0 \quad \text{as } \varepsilon \to 0.$$

- Convergence to the limit equation using a *perturbed test function* argument.
- *Uniqueness* of viscosity solution of the Hamilton-Jacobi equation thanks to:
  - constraint of negativity on u
  - $\max a$ attained in a unique point

Section 3

Biological Application

Oscillations on the optimal trait

- Periodic growth rate:

$$a_1(t,x) = r - g(x - c\sin bt)^2,\ \ r, g, c, b \in \mathbb{R}_+.$$

- Phenotypic mean of the population evolved in a periodic environment:

$$\mu_p(t) = \frac{1}{\rho_\varepsilon(t)} \int_{\mathbb{R}^d} x\, n_\varepsilon(t,x)\, dx \approx \frac{2\varepsilon c}{b}\sqrt{g}\sin\left(bt - \frac{\pi}{2}\right)$$

- The phenotypic mean trait oscillates with the same period of the growth rate function but experiments a "delay" with respect to the optimal trait.

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Load in the phenotypic mean

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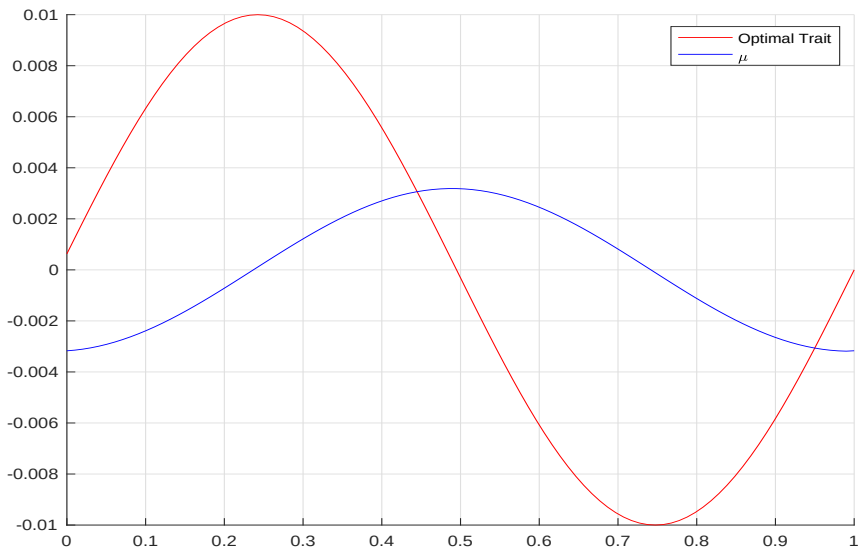

Figure 2: Comparison between the optimal trait and the phenotipycal mean (delay=1/4). ($r = 2, g = c = 1, b = 2\pi$)

Oscillations on the pressure of the selection

- Periodic growth rate:

$$a_2(t,x) = r - g(t)x^2, \quad r > 0, \quad g(t) : 1 - \text{periodic}.$$

- Variance and fitness for population evolved in periodic environment:

$$\sigma_p^2 \approx \frac{\varepsilon}{\sqrt{\bar{g}}}, \quad F_p(\tau) \approx r - \varepsilon \frac{g(1/2)}{\sqrt{\bar{g}}}.$$

- Variance and fitness for population evolved in constant environment ($t = 1/2$):

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Conclusions and future work

What we have done

- In long time, the solution converges to the unique periodic solution of the problem.
- When mutations are small the solution concentrates on a single Dirac mass, while the size of the population varies periodically in time.
- We obtain good approximations for the distribution's moments of the population.

What we are doing and perspectives

- Study of the impact of a climate shift: $a(e(t), x - ct)$.
- Rigorous computation of next order terms: $v(t, x)$.
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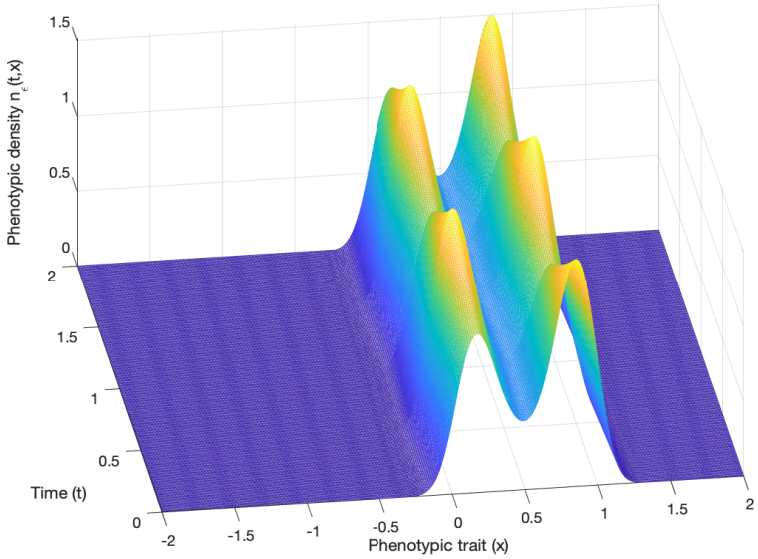
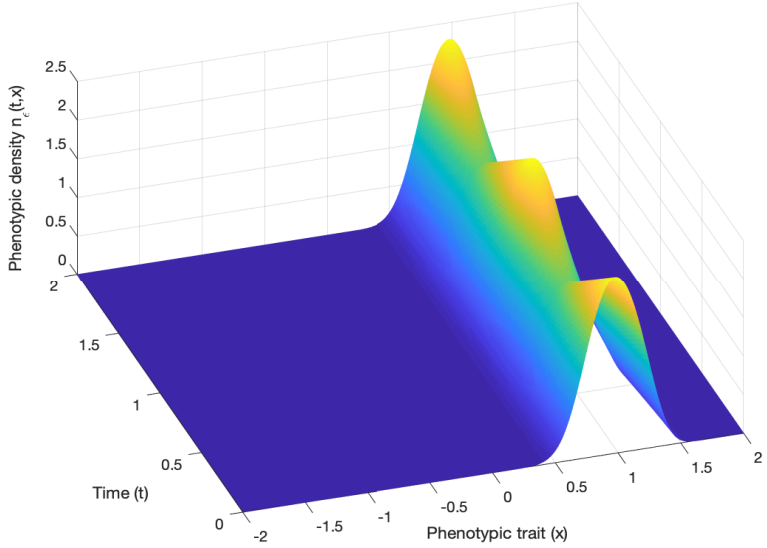
Several maxima case

$$a_3(t,x) = \begin{cases} R(x)(1-x), & \text{if } t \in (0,1/2), \\ R(x)x, & \text{if } t \in (1/2,1). \end{cases}$$

$$R_S(x) = -(x-0.1)^2(x-0.9)^2 \qquad R_{NS}(x) = -(x+1)^2(x-1)^4$$

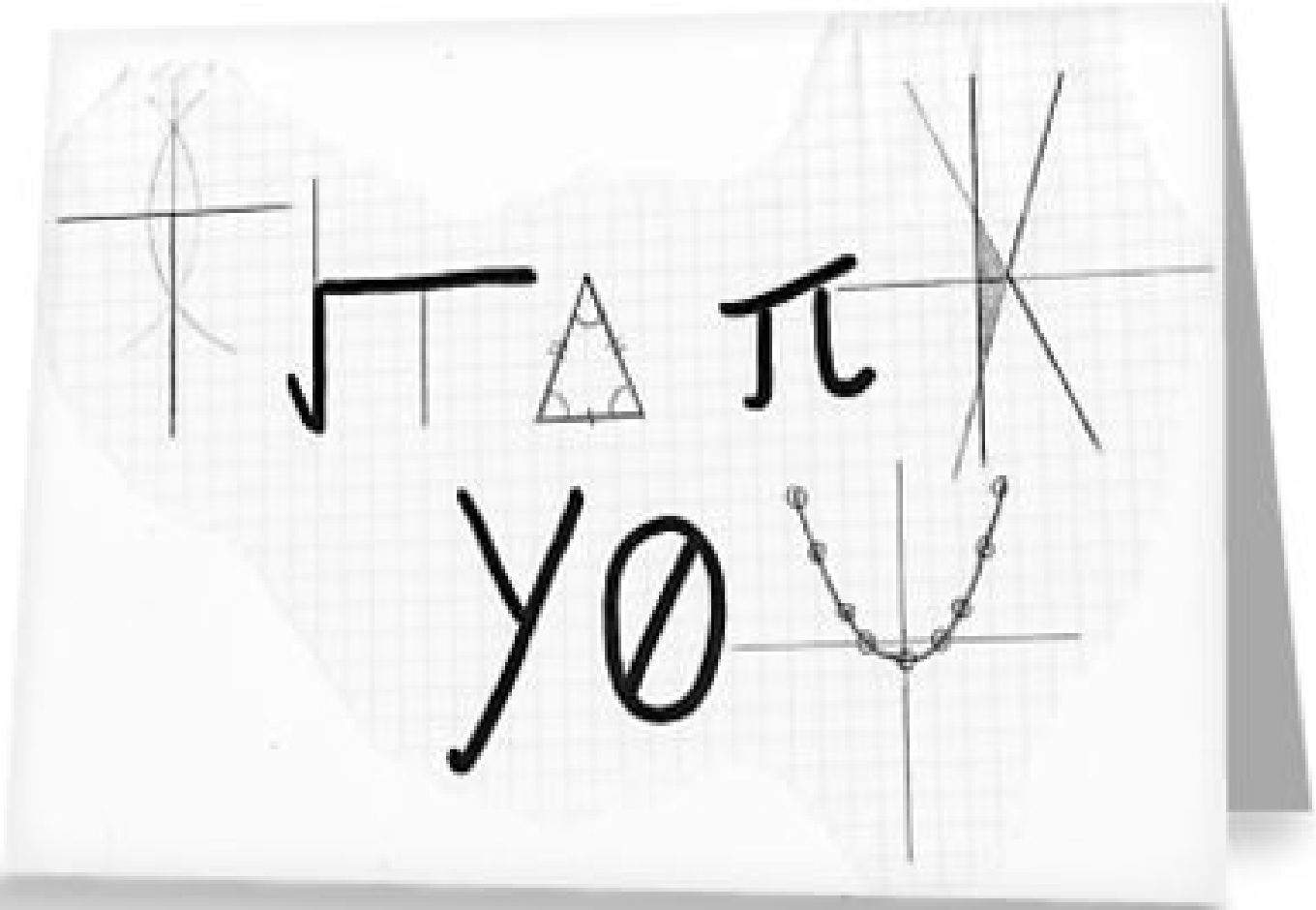


Dimorphic population



Monomorphic population

Figure 3: Phenotypic densities of the population.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